

**Figure 8.2** Georges Lemaître. Copyright Archives Georges Lemaître, Louvain.

The same thing happened again. Einstein had introduced the cosmological constant, the small but important modification of his equations I described in Chapter 3, in the (mistaken) hope of rendering the equations compatible with a static universe. When he had to acknowledge that the universe is not static, he turned against the cosmological constant. Lemaître, for the second time, tried to persuade him to change his mind: the cosmological constant does not render the universe static, but it is nevertheless right, and there is no reason to take it out. On this occasion, too, Lemaître was correct: the cosmological constant produces an acceleration of the expansion of the universe, and this acceleration has recently been measured. Once again, Einstein was wrong and Lemaître was right.

When the idea that the universe had emerged from a Big Bang began to be accepted, Pope Pius XII declared in a public address (on 22 November 1951) that the theory confirmed the account of Creation given in Genesis.[1] Lemaître reacted to this papal position with great concern. He got in touch with the scientific advisor to the pontiff and went to great lengths to persuade the Pope to refrain from making references to links between divine creation and the Big Bang. Lemaître was convinced that it was foolish to mix science and religion in this way: the Bible knows nothing about physics, and physics knows nothing about God.[2] Pius XII allowed himself to be persuaded, and the Catholic church never again made public allusion to the subject. It is not given to everyone to disprove the Pope.

And of course, on this also, it was Lemaître who was right: today there is a great deal of talk concerning the possibility that the Big Bang is not a real beginning, that there could have been another universe before it. Imagine in what an embarrassing position the Catholic Church would find itself today, if Lemaître had not prevented the Pope from making it official doctrine that that